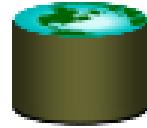


Constraints

Suppose that each of the following update operations is applied directly to the database state shown in Figure 5.6. Discuss *all* integrity constraints violated by each operation, if any, and the different ways of enforcing these constraints.

- a. Insert <'Robert', 'F', 'Scott', '943775543', '1952-06-21', '2365 Newcastle Rd, Bellaire, TX', M, 58000, '888665555', 1> into EMPLOYEE.
- b. Insert <'ProductA', 4, 'Bellaire', 2> into PROJECT.
- c. Insert <'Production', 4, '943775543', '1998-10-01'> into DEPARTMENT.
- d. Insert <'677678989', null, '40.0'> into WORKS_ON.
- e. Insert <'453453453', 'John', M, '1970-12-12', 'SPOUSE'> into DEPENDENT.
- f. Delete the WORKS_ON tuples with ESSN = '333445555'.
- g. Delete the EMPLOYEE tuple with SSN = '987654321'.

SAILOR DATABASE



Example Instances

R1

<u>sid</u>	<u>bid</u>	day
22	101	10/10/96
58	103	11/12/96

S1

<u>sid</u>	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S2

<u>sid</u>	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

<u>bid</u>	bname	color	
101	Interlake	blue	
102	Interlake	red	
103	Clipper	green	
104	Marine	red	

Boats



Projection

sid	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

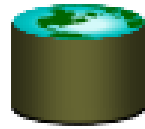
S2

sname	rating
yuppy	9
lubber	8
guppy	5
rusty	10

$$\pi_{sname, rating}(S2)$$

age
35.0
55.5

$$\pi_{age}(S2)$$



Selection (σ)

- Selects rows that satisfy *selection condition*.
- Result is a relation.
Schema of result is same as that of the input relation.
- Do we need to do duplicate elimination?

sid	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

$\sigma_{rating > 8}(S2)$

sname	rating
yuppy	9
rusty	10

$\pi_{sname, rating}(\sigma_{rating > 8}(S2))$



Union

<u>sid</u>	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S1

<u>sid</u>	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

S2

<u>sid</u>	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0
44	guppy	5	35.0
28	yuppy	9	35.0

$S1 \cup S2$



Set Difference

<u>sid</u>	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S1

<u>sid</u>	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

S2

<u>sid</u>	sname	rating	age
22	dustin	7	45.0

$S1 - S2$

<u>sid</u>	sname	rating	age
28	yuppy	9	35.0
44	guppy	5	35.0

$S2 - S1$



Cross Product Example

<u>sid</u>	<u>bid</u>	<u>day</u>
22	101	10/10/96
58	103	11/12/96

R1

<u>sid</u>	<u>sname</u>	<u>rating</u>	<u>age</u>
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S1

R1 X S1 =

(sid)	sname	rating	age	(sid)	bid	day
22	dustin	7	45.0	22	101	10/10/96
22	dustin	7	45.0	58	103	11/12/96
31	lubber	8	55.5	22	101	10/10/96
31	lubber	8	55.5	58	103	11/12/96
58	rusty	10	35.0	22	101	10/10/96
58	rusty	10	35.0	58	103	11/12/96



Intersection

<u>sid</u>	sname	rating	age
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S1

<u>sid</u>	sname	rating	age
28	yuppy	9	35.0
31	lubber	8	55.5
44	guppy	5	35.0
58	rusty	10	35.0

S2

<u>sid</u>	sname	rating	age
31	lubber	8	55.5
58	rusty	10	35.0

$$S1 \cap S2$$



Natural Join Example

<u>sid</u>	<u>bid</u>	<u>day</u>
22	101	10/10/96
58	103	11/12/96

R1

<u>sid</u>	<u>sname</u>	<u>rating</u>	<u>age</u>
22	dustin	7	45.0
31	lubber	8	55.5
58	rusty	10	35.0

S1

R1 * S1 =

<u>sid</u>	<u>sname</u>	<u>rating</u>	<u>age</u>	<u>bid</u>	<u>day</u>
22	dustin	7	45.0	101	10/10/96
58	rusty	10	35.0	103	11/12/96

Queries

- Find names of sailors who've reserved boat #103.
- Find names of sailors who have ratings at least 8.